Lecture 6: Dependent Sources and the Ideal Transformer

EE201 – Circuit Theory I

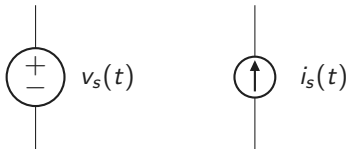
M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage power magnetic flux

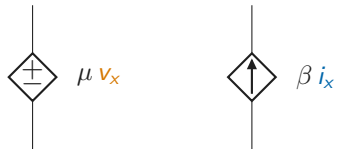
Sources controlled from elsewhere

independent



decides
its own value

dependent



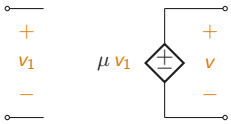
copies a variable
from elsewhere, scaled

Dependent (controlled) source

A voltage or current source whose value is a function of a voltage or current in *another* branch — the controlling variable. Almost always the function is a constant times one variable: a linear dependent source.

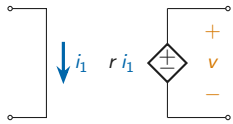
The four kinds

1. VCVS — voltage-controlled voltage source



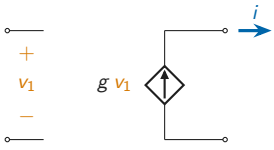
$v = \mu\, v_1$

2. CCVS — current-controlled voltage source



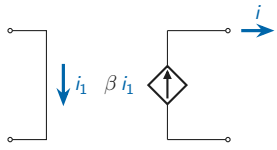
$v = r\, i_1$

3. VCCS — voltage-controlled current source



$i = g\, v_1$

4. CCCS — current-controlled current source



$i = \beta\, i_1$

The name reads in order: what it senses, then what it produces.

The constant, and its unit

Kind	Equation	The constant	Unit
VCVS	$v = \mu\, v_1$	voltage gain	none (V/V)
CCVS	$v = r\, i_1$	transresistance	ohm, Ω (V/A)
VCCS	$i = g\, v_1$	transconductance	siemens, S (A/V)
CCCS	$i = \beta\, i_1$	current gain	none (A/A)

The unit is always what comes out over what goes in.

Trans- when the two differ: a transresistance turns a current into a voltage, so it is measured in ohms.

More general forms

Several variables: $\mu_a v_a + \mu_b v_b$ — two dependent sources in series, one for each term.

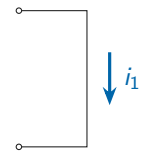
Affine: $\mu v_x + V_0$ — plus an independent source V_0.

Nonlinear: $i = f(v_x)$ — not used in EE201.

The sensing side



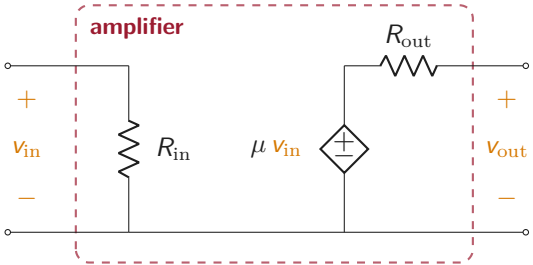
senses a voltage:
open circuit, $i = 0$



senses a current:
short circuit, $v = 0$

Sensing does not change — does not "disturb" — the circuit:
no current is drawn from a voltage, no voltage is taken from a current.

Why we need them

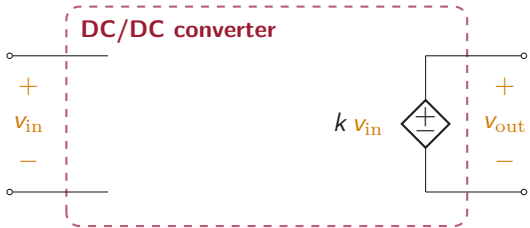


R_{in}: input resistance. R_{out}: output resistance.

The simplest model of anything that amplifies. Many components and integrated circuits are modelled as dependent sources — and some are designed to be one.

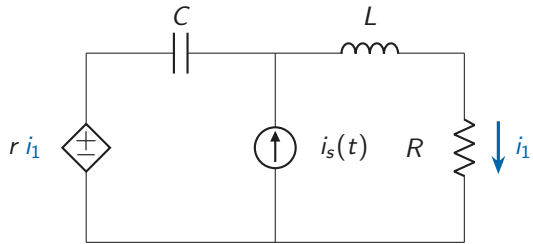
VCVS	op-amp	$\mu \approx 10^5$
CCVS	current-sense amplifier	r (Ω)
VCCS	MOSFET, transconductance amp.	g_m (S)
CCCS	bipolar transistor	$\beta \approx 100$

DC/DC converters



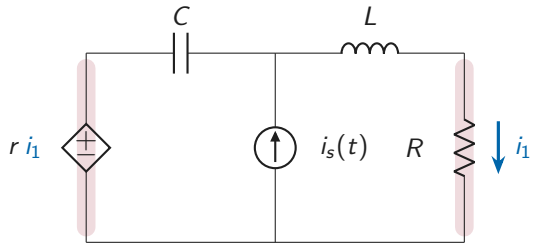
- They turn one DC voltage into another — USB $5\,\text{V}$ into the $1\,\text{V}$ a processor needs, a battery voltage into what a motor needs.
- One of the most fundamental electronic circuits, in almost every field: phones, computers, cars, satellites, robots.
- To the circuit around it, a DC/DC converter is essentially a VCVS: $v_{\text{out}} = k\,v_{\text{in}}$, with k set by its design.

Example: the controlling variable of a CCVS



Which equation does the dependent source contribute?

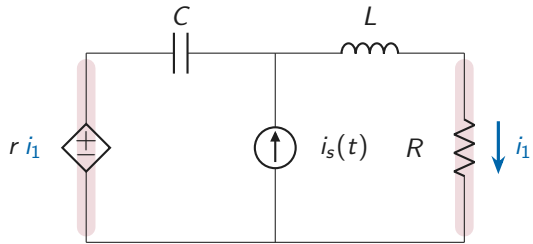
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Which equation does the dependent source contribute?

A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

Example: the controlling variable of a CCVS

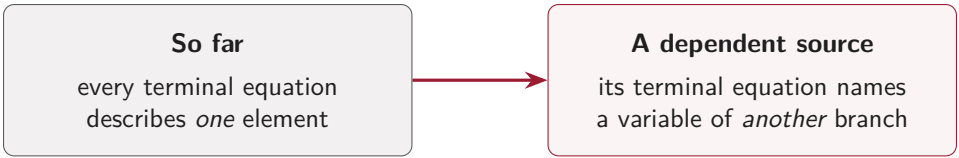


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A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

$n = 4$ nodes and $b = 5$ branches: $2b = 10$ unknowns, $b = 5$ terminal equations.
Four speak about one element each. The fifth links two branches.

Terminal equations with a dependent source



KCL and KVL are unchanged. Still $2b$ equations for $2b$ unknowns.

Two meanings of "controlled"

Earlier: a *voltage-controlled resistor* has $i = f(v)$ of its own voltage. A voltage-controlled *source* takes its value from another branch.

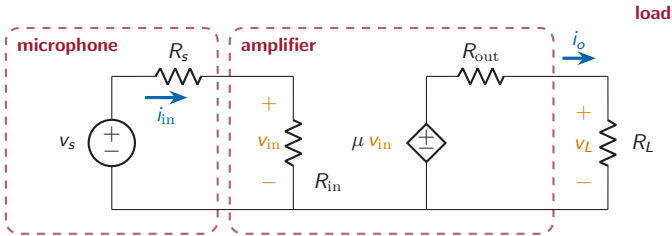
Solving circuits with dependent sources

Recipe

1. Treat the dependent source as if it were independent, with its value written as μv_x, $r i_x$, $g v_x$ or βi_x.
2. Write KCL, KVL and the terminal equations as usual.
3. Add one equation that gives the controlling variable in terms of your unknowns.

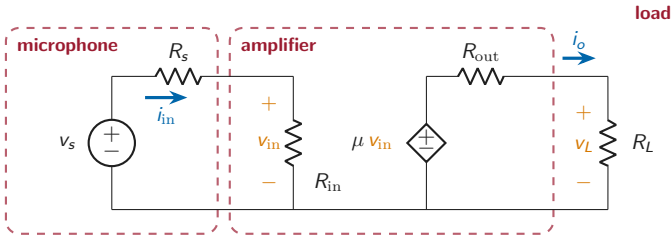
The power check $\sum_k p_k = 0$ still works — include the dependent source.

Example: an amplifier — find v_L



$v_s = 50\,\text{mV}, \;\; R_s = 1\,\text{k}\Omega \quad | \quad R_{\text{in}} = 4\,\text{k}\Omega, \;\; \mu = 100, \;\; R_{\text{out}} = 100\,\Omega \quad | \quad R_L = 400\,\Omega$

Example: an amplifier — find v_L



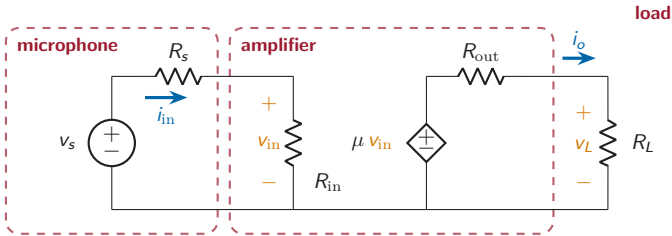
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Input side — a divider:

$$i_{\text{in}} = \frac{50\,\text{mV}}{5\,\text{k}\Omega} = 10\,\mu\text{A}$$

$$v_{\text{in}} = 4\,\text{k}\Omega \times 10\,\mu\text{A} = 40\,\text{mV}$$

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Output side — the source gives 4 V:

$$i_o = \frac{4\,\text{V}}{500\,\Omega} = 8\,\text{mA}$$

$$v_L = 3.2\,\text{V}, \qquad \frac{v_L}{v_s} = 64$$

Where the power comes from

input side	
microphone	$-0.5\,\mu\mathrm{W}$
R_s	$+0.1\,\mu\mathrm{W}$
R_{in}	$+0.4\,\mu\mathrm{W}$

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output side	
dependent source	$-32\,\mathrm{mW}$
R_{out}	$+6.4\,\mathrm{mW}$
R_L	$+25.6\,\mathrm{mW}$

Each side sums to zero.

Where the power comes from

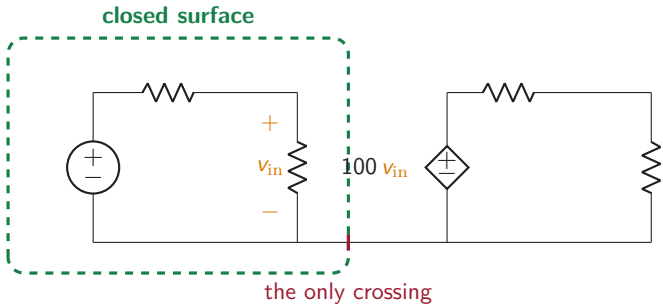
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Each side sums to zero.

The load gets 51 200 times what the microphone supplies.
The dependent source delivered it. A real amplifier takes it from its power supply, which the model does not draw.

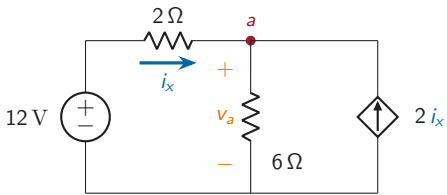
The shared wire carries no current



KCL on the closed surface: one branch crosses it, so its current is zero.

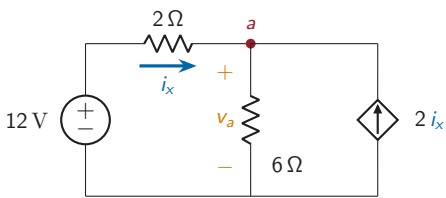
The two sides interact only through v_{in}.

Example: when the controlling variable is unknown



Find i_x and v_a.

Example: when the controlling variable is unknown



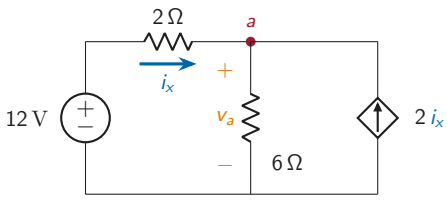
Find i_x and v_a.

Equations — i_x is already an unknown:

$$\text{KVL:}\quad -12 + 2i_x + v_a = 0$$

$$\text{KCL at } a\text{:}\quad -i_x - 2i_x + \tfrac{v_a}{6} = 0 \;\Rightarrow\; v_a = 18\, i_x$$

Example: when the controlling variable is unknown



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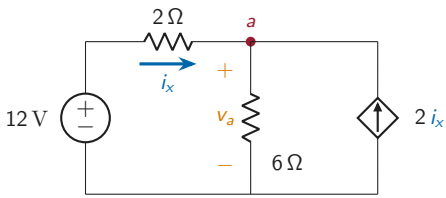
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Solution: $-12 + 20\, i_x = 0$

$$i_x = 0.6\,\text{A}, \qquad v_a = 10.8\,\text{V}$$

Example: when the controlling variable is unknown



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Power check:

$$\underbrace{-7.2}_{12\,\text{V}} + \underbrace{0.72}_{2\Omega} + \underbrace{19.44}_{6\Omega} \underbrace{-\,12.96}_{\text{CCCS}} = 0$$

The CCCS carries $1.2\,\text{A}$ *out of* its + terminal: it delivers.

What a dependent source cannot do

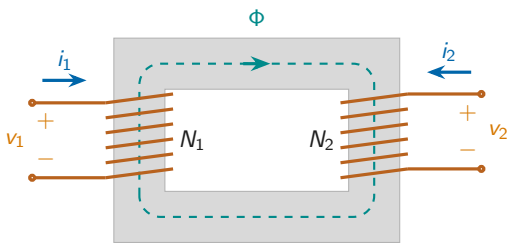
It cannot start anything

Switch the $12\,\mathrm{V}$ off: $2i_x + v_a = 0$ and $v_a = 18i_x$, so $20\,i_x = 0$.
Every voltage and current is zero — the dependent source's too.

It is never switched off

Later we switch sources off one at a time (superposition, Thévenin). Only independent sources are switched off: a dependent source is part of how the circuit behaves, not an input.

The transformer

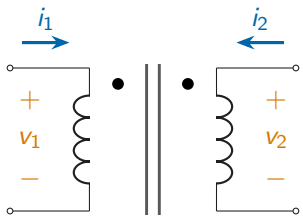


- Sometimes we need a magnetic link between two circuits. We model it as an element: the transformer.
- A current in coil 1 sets up a magnetic field; the iron core guides it through coil 2.
- When the current changes, the field changes and induces a voltage in coil 2. No wire connects the coils.

A physical transformer works only with AC signals

A constant current gives a constant field, and no voltage appears on the other coil.

The ideal transformer



Two equations

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} \qquad N_1 i_1 + N_2 i_2 = 0$$

v positive and i entering at the dots.

Move one dot and both equations change sign.

No d/dt: in the model it is a resistive element.

For simplicity, the ideal model also works for DC.

The ideal transformer absorbs no power

$$p(t) = v_1 i_1 + v_2 i_2 = 0 \qquad \text{at every instant}$$

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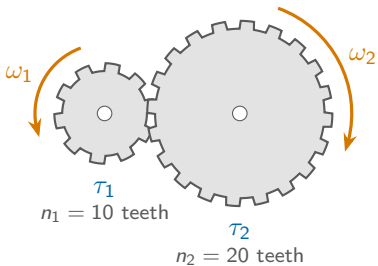
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Voltage multiplied by N_2/N_1, current divided by N_2/N_1.

Nothing is stored, nothing is dissipated. The same holds for any number of windings.

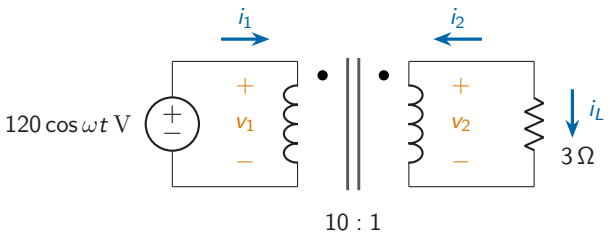
A mechanical analogy: the gear pair



Ideal transformer	**Ideal gear pair**
voltage v	angular velocity ω
current i	torque τ
turns N_1, N_2	teeth n_1, n_2
power in = power out	$\tau_1\omega_1 = \tau_2\omega_2$
step-down: v down, i up	ω down, τ up
dots set the sign	gears turn opposite ways

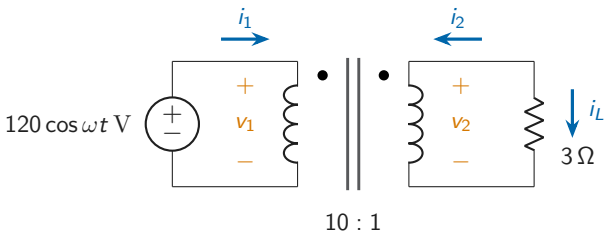
10 teeth driving 20: half the speed, twice the torque — like a $2:1$ step-down transformer.

Example: a step-down transformer



Find the currents and the power on each side.

Example: a step-down transformer



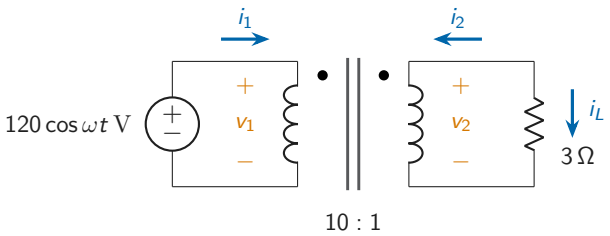
Find the currents and the power on each side.

$v_2 = \frac{1}{10} v_1 = 12\cos\omega t$ V

$i_L = v_2/3 = 4\cos\omega t$ A, so $i_2 = -4\cos\omega t$ A

$i_1 = -\frac{1}{10} i_2 = 0.4\cos\omega t$ A

Example: a step-down transformer



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$i_1 = -\frac{1}{10} i_2 = 0.4\cos\omega t$ A

$p_1 = v_1 i_1 = 48\cos^2\omega t$ W

$p_2 = v_2 i_2 = -48\cos^2\omega t$ W

Seen from the source:
$v_1/i_1 = 300\,\Omega = 10^2 \times 3\,\Omega$.

Summary

- A **dependent source** (diamond) takes its value from a voltage or current in another branch: VCVS, CCVS, VCCS, CCCS. Unit of the constant: out over in. Almost always linear in one variable.
- Its terminal equation names a foreign variable. Write KCL, KVL and terminal equations as usual, plus one equation for the controlling variable.
- It can **deliver power** — that is amplification — but it **cannot start anything**, and it is never switched off.
- **Ideal transformer**: $v_1/N_1 = v_2/N_2$, $N_1 i_1 + N_2 i_2 = 0$, signs set by the dots. It absorbs **no power at any instant**. A physical one works only with AC signals.

Next lecture

From circuit to graph

Labelling the nodes and branches of a circuit so that its $2b$ equations can be written systematically — the first step towards node and mesh analysis.

Questions?